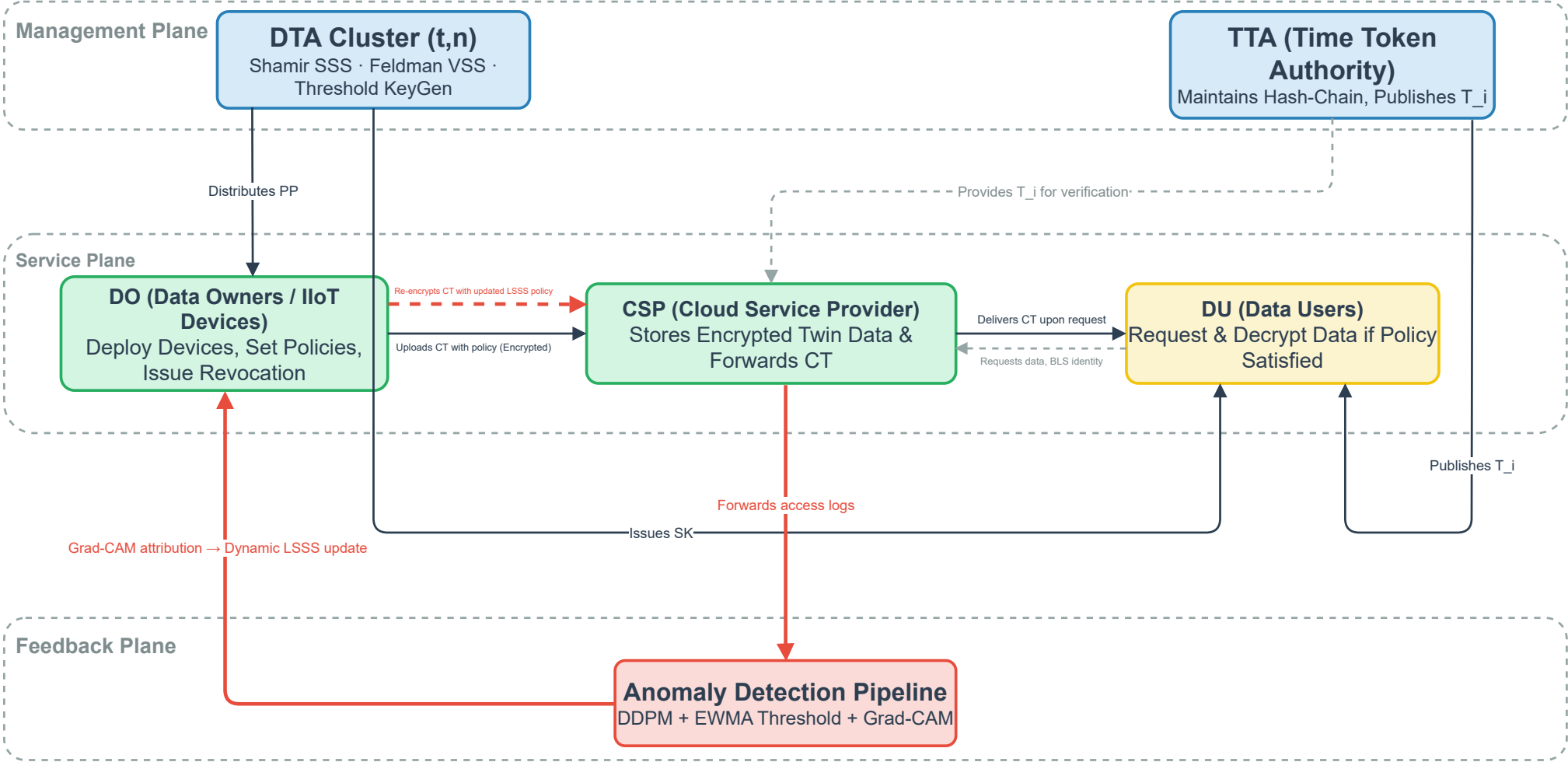


Fig. 1. System architecture of Sentinel-CP-ABE showing the DTA cluster, TTA, five service entities (DO, CSP, DU), and the AI-driven closed-loop anomaly detection pipeline with Grad-CAM attribution.



Data Flow: DO → CSP → DU | **Key Dist.:** DTA/TTA → DO/DU
AI Feedback: CSP → DDPM+EWMA → Grad-CAM → DO → CSP (Re-encrypt)

Legend

- Data / Key distribution
 - Request / Verification
 - Anomaly-driven feedback
- Mgt / Svc / Fbk